

Triangle congruence tests



1

- a Yes, they satisfy RHS.
- b Unknown, there is not enough information.
- c Yes, they satisfy AAS.
- d No, they are definitely not congruent.
- e Yes, they satisfy SAS.
- f Yes, they satisfy RHS.
- g No, they are definitely not congruent.
- h Yes, they satisfy SSS.
- i Yes, they satisfy AAS.
- j Yes, they satisfy SSS.

2

- a Triangle C and D
- b AAS

3

- a Triangle B and C
- b RHS

4

- a Triangle C and D
- b SSS

5

- a Triangle A and B
- b AAS

6

- a Triangle B and C
- b SAS

7

- a Triangle A and C
- b RHS

8

- a Triangle A and C
- b SAS

9

- a Triangle B and C
- b RHS

10



11 SAS

12

a $\triangle ACD$ and $\triangle BCE$

b SSS

13

a $\triangle JKL$ and $\triangle JMN$

b AAS

14

a $\triangle PRN$ and $\triangle QRM$.

b SSS and SAS.

15 Possible Answers:

$$\angle RPQ = \angle RPS, QR = QS, \angle QRP = \angle SRP$$

16 Possible Answers:

$$JN = MN, \angle JKN = \angle MLN, \angle KJN = \angle LMN$$

17 Possible Answers:

$$\angle PXQ = \angle RXS, \angle PQX = \angle SRX, \angle PQX = \angle RSX, \angle QPX = \angle SRX, \\ \angle QPX = \angle RSX, PQ = SR$$

18

a SAS

b $EF = EH$ Given

$\angle FEG = \angle HED$ Vertically opposite angles

$EG = ED$ Given

19

a SAS

b $\angle AXB = \angle DXC$ Vertically opposite angles

$BX = CX$ Radii of the same circle

$AX = DX$ Radii of the same circle

20

a SAS

b $AB = CD$ Given



BD is common

$\angle ABD = \angle CDB$ Alternate angles on parallel lines

21

a AAS or SAS

b Example Answer:

$BX = CX$ Given

$\angle ABX = \angle DCX$ Alternate angles on parallel lines

$\angle BXA = \angle CXD$ Vertically opposite angles